

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Google Associate Cloud Engineer — Exam Overview

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

About the Certification

The **Google Associate Cloud Engineer** certification demonstrates your ability to deploy and secure applications and infrastructure, monitor operations of multiple projects, and maintain enterprise solutions to ensure they meet target performance metrics. This is a foundational-level certification ideal for cloud professionals who want to validate their GCP skills.

Exam Details

Item	Detail
Exam length	2 hours (120 minutes)
Question count	50–60 questions
Question format	Multiple choice and multiple select
Passing score	~70 % (Google does not publish exact cut-off)
Registration fee	\$200 USD
Languages	English, Japanese, Spanish, Portuguese, French, German, Indonesian, Korean, Chinese
Recommended experience	6+ months hands-on with GCP
Certification validity	2 years (recertification required)
Delivery	Online proctored or test center (Kryterion)

Exam Sections and Weights

Section	Weight
1. Setting up a cloud solution environment	~20 %
2. Planning and configuring a cloud solution	~17.5 %
3. Deploying and implementing a cloud solution	~25 %
4. Ensuring successful operation of a cloud solution	~20 %
5. Configuring access and security	~17.5 %

Section 1 — Setting Up a Cloud Solution Environment (~20 %)

1.1 Setting up cloud projects and accounts

- Creating a resource hierarchy
- Applying organizational policies to the resource hierarchy
- Granting members IAM roles within a project
- Managing users and groups in Cloud Identity (manually and automated)
- Enabling APIs within projects
- Provisioning and setting up products in Google Cloud's operations suite
- Assessing quotas and requesting increases

1.2 Managing billing configuration

- Creating one or more billing accounts
 - Linking projects to a billing account
 - Establishing billing budgets and alerts
 - Setting up billing exports
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Section 2 — Planning and Configuring a Cloud Solution (~17.5 %)

2.1 Planning and configuring compute resources

- Selecting appropriate compute choices for a given workload (Compute Engine, GKE, Cloud Run, Cloud Functions)
- Using Spot VM instances and custom machine types as appropriate

2.2 Planning and configuring data storage options

- Product choice (Cloud SQL, BigQuery, Firestore, Spanner, Bigtable)
- Choosing storage options (Zonal Persistent Disk, Regional Persistent Disk, Standard, Nearline, Coldline, Archive)

2.3 Planning and configuring network resources

- Load balancing
 - Availability of resource locations in a network
 - Network Service Tiers
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Section 3 — Deploying and Implementing a Cloud Solution (~25 %)

3.1 Deploying and implementing Compute Engine resources

- Launching a compute instance (assign disks, availability policy, SSH keys)
- Creating an autoscaled managed instance group by using an instance template
- Configuring OS Login
- Configuring VM Manager

3.2 Deploying and implementing Google Kubernetes Engine resources

- Installing and configuring kubectl
- Deploying a GKE cluster (Autopilot, regional clusters, private clusters, GKE Enterprise)
- Deploying a containerized application to GKE

3.3 Deploying and implementing Cloud Run and Cloud Functions resources

- Deploying an application
- Deploying an application for receiving Google Cloud events (Pub/Sub, Cloud Storage, Eventarc)
- Determining where to deploy (Cloud Run fully managed, Cloud Run for Anthos, Cloud Functions)

3.4 Deploying and implementing data solutions

- Deploying data products (Cloud SQL, Firestore, BigQuery, Spanner, Pub/Sub, Dataflow, Cloud Storage, AlloyDB)
- Loading data (command line upload, from Cloud Storage, Storage Transfer Service)

3.5 Deploying and implementing networking resources

- Creating a VPC with subnets (custom mode VPC, Shared VPC)
- Creating ingress and egress firewall rules (IP subnets, network tags, service accounts)
- Peering external networks (Cloud VPN, VPC Network Peering)

3.6 Implementing resources through infrastructure as code

- Infrastructure as code tooling (Cloud Foundation Toolkit, Config Connector, Terraform, Helm)
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Section 4 — Ensuring Successful Operation of a Cloud Solution (~20

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4.1 Managing Compute Engine resources

- Remotely connecting to instances
- Viewing current running VM inventory (instance IDs, details)
- Working with snapshots (create, view, delete, schedule)
- Working with images (create from VM or snapshot, view, delete)

4.2 Managing Google Kubernetes Engine resources

- Viewing current running cluster inventory (nodes, pods, services)
- Configuring GKE to access Artifact Registry
- Working with node pools (add, edit, remove)
- Working with Kubernetes resources (pods, services, statefulsets)
- Managing horizontal and vertical autoscaling configurations

4.3 Managing Cloud Run resources

- Deploying new versions of an application
- Adjusting application traffic splitting parameters
- Setting scaling parameters for autoscaling instances

4.4 Managing storage and database solutions

- Managing and securing objects in Cloud Storage buckets
- Setting object lifecycle management policies
- Executing queries (Cloud SQL, BigQuery, Spanner, Firestore, AlloyDB)
- Estimating costs of data storage resources
- Backing up and restoring database instances (Cloud SQL, Firestore)
- Reviewing job status (Dataflow, BigQuery)

4.5 Managing networking resources

- Adding a subnet to an existing VPC
- Expanding a subnet to have more IP addresses
- Reserving static external or internal IP addresses
- Working with Cloud DNS and Cloud NAT

4.6 Monitoring and logging

- Creating Cloud Monitoring alerts based on resource metrics

- Creating and ingesting custom metrics (from applications or logs)
 - Exporting logs to external systems (on-premises, BigQuery)
 - Configuring log buckets, log analytics, and log routers
 - Viewing and filtering logs in Cloud Logging
 - Viewing specific log message details
 - Using cloud diagnostics to research application issues
 - Viewing Google Cloud status
 - Configuring and deploying Ops Agent
 - Deploying Managed Service for Prometheus
 - Configuring audit logs
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Section 5 — Configuring Access and Security (~17.5 %)

5.1 Managing Identity and Access Management (IAM)

- Viewing and creating IAM policies
- Managing role types and defining custom IAM roles (basic, predefined, custom)

5.2 Managing service accounts

- Creating service accounts
 - Using service accounts in IAM policies with minimum permissions
 - Assigning service accounts to resources
 - Managing IAM of a service account
 - Managing service account impersonation
 - Creating and managing short-lived service account credentials
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Study Strategy

Recommended Resources

1. **Google Cloud Skills Boost** — Free labs and quests (cloud.google.com/training)
2. **Official Documentation** — cloud.google.com/docs
3. **Coursera: Google Cloud Fundamentals** — Core Infrastructure course
4. **A Cloud Guru / Pluralsight** — Dedicated ACE courses

5. **Qwiklabs** — Hands-on labs for each service
6. **Practice Exams** — Google offers a free practice exam

Study Tips

- **Hands-on practice is critical** — The exam is practical, not theoretical
- **Know the gcloud CLI** — Many questions test command-line knowledge
- **Understand pricing** — Know when to use preemptible/spot VMs, committed use discounts
- **Focus on Section 3** — It has the highest weight (25 %) and is considered the hardest
- **Learn IAM deeply** — Roles, service accounts, and policies appear throughout the exam
- **Know when to use each service** — Compute Engine vs GKE vs Cloud Run vs Cloud Functions
- **Understand networking basics** — VPCs, subnets, firewall rules, load balancing
- **Time management** — ~2 minutes per question; flag and revisit difficult ones

Key gcloud Commands to Memorize

```
gcloud config set project PROJECT_ID
gcloud compute instances list
gcloud compute instances create INSTANCE_NAME --zone=ZONE
gcloud container clusters create CLUSTER_NAME
gcloud run deploy SERVICE_NAME --image IMAGE
gcloud functions deploy FUNCTION_NAME
gcloud iam roles create ROLE_ID
gcloud projects add-iam-policy-binding PROJECT_ID --member=MEMBER --role=ROLE
```

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<code>02-managing-billing-configuration.md</code>	Section 1.2 — Billing accounts, budgets, alerts, exports
<code>03-planning-compute-resources.md</code>	Section 2.1 — Compute Engine, GKE, Cloud Run, Cloud Functions
<code>04-planning-data-storage-options.md</code>	Section 2.2 — Cloud SQL, BigQuery, Firestore, Spanner, Bigtable, Storage classes
<code>05-planning-network-resources.md</code>	Section 2.3 — Load balancing, locations, Network Service Tiers
<code>06-deploying-compute-engine.md</code>	Section 3.1 — Launching VMs, MIGs, OS Login, VM Manager
<code>07-deploying-gke-resources.md</code>	Section 3.2 — kubectl, GKE clusters, containerized apps
<code>08-deploying-cloud-run-and-functions.md</code>	Section 3.3 — Cloud Run, Cloud Functions, Eventarc
<code>09-deploying-data-solutions.md</code>	Section 3.4 — Data products, loading data
<code>10-deploying-networking-resources.md</code>	Section 3.5 — VPCs, firewall rules, VPN, peering
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<code>12-managing-compute-engine.md</code>	Section 4.1 — VM management, snapshots, images
<code>13-managing-gke-resources.md</code>	Section 4.2 — Cluster management, node pools, autoscaling
<code>14-managing-cloud-run.md</code>	Section 4.3 — Versioning, traffic splitting, scaling
<code>15-managing-storage-and-databases.md</code>	Section 4.4 — Object management, lifecycle, queries, backups
<code>16-managing-networking-resources.md</code>	Section 4.5 — Subnets, IPs, Cloud DNS, Cloud NAT

File	Topic
17-monitoring-and-logging.md	Section 4.6 — Monitoring, logging, alerts, Ops Agent
18-managing-iam.md	Section 5.1 — IAM policies, role types, custom roles
19-managing-service-accounts.md	Section 5.2 — Service accounts, impersonation, credentials

Glossary

ACE (Associate Cloud Engineer) — Google Cloud certification that validates the ability to deploy, monitor, and manage cloud solutions on GCP.

AlloyDB — Fully managed, PostgreSQL-compatible database service offering high performance for both transactional and analytical workloads.

Anthos — Google's hybrid and multi-cloud application platform, enabling GKE workloads to run on-premises or across multiple clouds.

API (Application Programming Interface) — A programmatic interface that exposes a service's functionality; GCP services are consumed via APIs that must be explicitly enabled per project.

App Engine — Google Cloud's fully managed platform-as-a-service (PaaS) for deploying web applications without managing infrastructure.

Artifact Registry — Google Cloud's fully managed service for storing and managing container images, language packages, and other build artifacts.

Audit Logs — Records of administrative activity and data access events within GCP, used for security and compliance monitoring via Cloud Logging.

Autoscaler — GCP component that automatically adjusts the number of VM instances in a managed instance group based on defined metrics and policies.

Autopilot — A GKE mode where Google fully manages node provisioning and scaling; billing is per pod rather than per node.

Bigtable (Cloud Bigtable) — Fully managed, wide-column NoSQL database designed for massive analytical and operational workloads with millisecond latency.

BigQuery — Serverless, highly scalable data warehouse for analytical (OLAP) workloads, capable of processing petabytes of data using SQL.

BigQuery ML — Feature within BigQuery that allows creation and execution of machine learning models using SQL.

Cloud DNS — Google Cloud's scalable, reliable, and managed authoritative Domain Name System (DNS) service.

Cloud Foundation Toolkit (CFT) — A set of reference templates and best practices for deploying GCP infrastructure using Terraform and other IaC tools.

Cloud Functions — Serverless, event-driven Function as a Service (FaaS) platform where code runs in response to triggers without managing infrastructure.

Cloud Identity — Google's Identity as a Service (IDaaS) solution for managing users, groups, devices, and SSO across Google Cloud.

Cloud Logging — Centralized log management service that ingests, stores, and routes logs from GCP services and applications.

Cloud Monitoring — GCP service for collecting metrics, creating dashboards, setting alerts, and performing uptime checks on resources.

Cloud NAT — Managed network address translation service that allows VMs without external IPs to access the internet.

Cloud Profiler — Continuous CPU and memory profiling service for production applications running on GCP.

Cloud Run — Fully managed serverless platform for running containers, scaling automatically including to zero.

Cloud Run for Anthos — Cloud Run deployed on GKE clusters, enabling serverless container execution in hybrid environments.

Cloud Scheduler — Fully managed cron job service for scheduling tasks such as invoking Cloud Functions or HTTP endpoints.

Cloud SQL — Fully managed relational database service supporting MySQL, PostgreSQL, and SQL Server.

Cloud Storage — Google Cloud's object storage service for storing and retrieving unstructured data in buckets.

Cloud Trace — Distributed tracing service for analyzing request latency across applications on GCP.

Cloud VPN — Managed service for securely connecting on-premises networks or other cloud networks to GCP VPCs over IPsec tunnels.

CaaS (Container as a Service) — Cloud service model where the provider manages the container orchestration infrastructure; GKE is GCP's CaaS offering.

Committed Use Discounts (CUDs) — Discounts of up to 57% (1-year) or 70% (3-year) on Compute Engine and Cloud SQL in exchange for a resource usage commitment.

Compute Engine — GCP's Infrastructure as a Service (IaaS) offering providing virtual machines running on Google's infrastructure.

Config Connector — Kubernetes add-on that allows managing GCP resources using Kubernetes manifests and the Kubernetes API.

Dataflow — Fully managed stream and batch data processing service based on the Apache Beam programming model.

Eventarc — GCP service that enables event-driven architectures by routing events from GCP services and custom sources to Cloud Run, Cloud Functions, and other targets.

FaaS (Function as a Service) — Serverless compute model where individual functions are deployed and executed in response to events; Cloud Functions is GCP's FaaS offering.

Firestore — Serverless, NoSQL document database designed for mobile, web, and IoT applications with real-time sync and offline support.

GCP (Google Cloud Platform) — Google's suite of cloud computing services running on the same infrastructure that Google uses internally.

gcloud CLI — The primary command-line interface for interacting with GCP services and resources.

GKE (Google Kubernetes Engine) — Google Cloud's managed Kubernetes service for deploying and operating containerized applications.

GKE Enterprise — Enhanced tier of GKE that provides additional features for multi-cluster and hybrid deployments, formerly part of Anthos.

Helm — Package manager for Kubernetes that uses charts (pre-configured application definitions) to deploy and manage applications.

HPC (High-Performance Computing) — Workloads requiring significant computational resources, often using clusters of compute-optimized VMs or GPUs.

HTTP(S) Load Balancer — GCP's global Layer 7 load balancer that distributes HTTP and HTTPS traffic across backends.

IaaS (Infrastructure as a Service) — Cloud service model providing virtualized computing resources such as VMs, storage, and networking.

IAM (Identity and Access Management) — GCP's system for controlling who (members/principals) has what access (roles) to which resources.

Kryterion — Testing platform used to deliver Google Cloud certification exams at physical test centers.

kubectl — Command-line tool for interacting with Kubernetes clusters, used to deploy and manage applications on GKE.

Managed Instance Group (MIG) — A group of identical VM instances managed as a single entity, supporting autoscaling, auto-healing, and rolling updates.

Managed Service for Prometheus — Google Cloud's managed monitoring solution compatible with the Prometheus open-source monitoring system.

Ops Agent — Unified agent for Compute Engine VMs that collects logs and metrics and sends them to Cloud Logging and Cloud Monitoring.

Operations Suite — Formerly Stackdriver; Google Cloud's integrated suite of monitoring, logging, tracing, and diagnostics tools.

OS Login — Feature that allows managing SSH access to Compute Engine instances using Google account IAM roles instead of SSH keys.

PagerDuty — Third-party incident management platform that can receive Cloud Monitoring alert notifications.

Preemptible VM — Legacy low-cost VM type that Google can reclaim at any time with a 30-second warning and a maximum lifetime of 24 hours; superseded by Spot VMs.

Pub/Sub (Cloud Pub/Sub) — Fully managed asynchronous messaging service for decoupling applications using a publish-subscribe model.

Qwiklabs — Hands-on cloud learning platform providing lab environments for practicing GCP skills.

Service Account — Special Google account representing an application or workload rather than a human user, used for authenticating service-to-service calls.

Shared VPC — GCP feature allowing a host project to share its VPC network with service projects, centralizing network management.

Snapshot — Point-in-time copy of a Persistent Disk used for backup and disaster recovery.

Spanner (Cloud Spanner) — Globally distributed, strongly consistent, horizontally scalable relational database with 99.999% SLA.

Spot VM — Low-cost Compute Engine VM type offering up to 91% discount; Google can reclaim it at any time with 30-second notice and no maximum runtime limit.

Stackdriver — Former name of the Google Cloud Operations Suite (now Cloud Monitoring, Cloud Logging, Cloud Trace, etc.).

Statefulset — Kubernetes workload object for managing stateful applications, providing stable network identities and persistent storage.

Storage Transfer Service — GCP service for transferring large amounts of data from on-premises, other clouds, or URLs into Cloud Storage.

Sustained Use Discounts (SUDs) — Automatic Compute Engine discounts of up to 30% applied when a VM runs for more than 25% of a billing month.

Terraform — Open-source Infrastructure as Code tool by HashiCorp used to define and provision GCP resources declaratively.

TPU (Tensor Processing Unit) — Google-designed application-specific integrated circuit (ASIC) optimized for machine learning workloads, particularly TensorFlow and JAX.

VPC (Virtual Private Cloud) — Logically isolated network within GCP where you can launch and manage resources with custom IP ranges, subnets, and routing.

VPC Network Peering — Feature that connects two VPC networks so resources in each network can communicate using internal IP addresses.

VM (Virtual Machine) — Emulation of a physical computer running an operating system and applications; the core unit of Compute Engine.

VM Manager — Suite of tools for managing operating systems on Compute Engine VM fleets, including patch management and inventory.

WebSocket — Protocol providing full-duplex communication over a single TCP connection, supported by Cloud Run for real-time applications.